

Derivation v17: EW-Scale Calibration Robustness for R_ξ and M_5

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Abstract

This note stress-tests the Track B calibrated closure from v16 by replacing the single identification $R_\xi = \hbar c/M_Z^{\text{obs}}$ with a calibration family $R_\xi = \hbar c/M_*^{\text{obs}}$ where $M_* \in \{M_Z, M_W, v_{\text{EW}}\}$. We quantify how M_5 shifts under these alternatives and provide a robustness verdict. **Result:** All three EW scales yield M_5 within the same decade ($2.3\text{--}3.4 \times 10^{13}$ GeV), with maximum shift $|\Delta \log_{10} M_5| < 0.15$. The calibration is **robust**. We recommend M_Z as the canonical choice due to superior metrological precision and definitional stability.

1 Recap: v15 Calibrated Closure and v16 Track B

Derivation v15 established the calibrated closure formula for the 5D Planck mass:

$$M_5 = M_{\text{Pl}}^{2/3} R_\xi^{-1/3} \quad (1)$$

where $M_{\text{Pl}}^{\text{obs}} = 1.221 \times 10^{19}$ GeV is the observed 4D Planck mass [BL], and R_ξ is the compactification scale requiring specification.

Derivation v16 pursued two tracks:

- **Track A (internal derivation):** NO-GO — R_ξ cannot be derived from EDC axioms alone.
- **Track B (minimal baseline):** CLOSED — Identify $R_\xi = \hbar c/M_Z^{\text{obs}}$ with $M_Z^{\text{obs}} = 91.1876$ GeV, yielding $R_\xi = 2.165 \times 10^{-18}$ m and $M_5 = 2.4 \times 10^{13}$ GeV.

Epistemic status of Track B: The identification $R_\xi = \hbar c/M_Z$ is tagged as [I]+[BL] (identification plus baseline measurement), not [D] (derived). This is methodologically honest but raises the question: *Why M_Z and not another EW scale?*

2 Calibration Family Definition

We generalize the Track B identification to a **calibration family**:

$$R_\xi = \frac{\hbar c}{M_*^{\text{obs}}} \quad (2)$$

where M_* is an electroweak scale. The candidates considered are those already referenced in the EDC repository:

1. $M_Z = 91.1876 \pm 0.0021$ **GeV** — Z boson pole mass [BL]. Primary reference in derivation notes v2–v16 and Part II Ch. 3.
2. $M_W = 80.379 \pm 0.012$ **GeV** — W boson pole mass [Dc]. Derived in EDC via $M_W = gv_{\text{EW}}/2$ with 0.2% agreement to experiment.

3. $v_{\text{EW}} = 246.22 \pm 0.01$ **GeV** — Electroweak vacuum expectation value [BL]. Defined via $v = (\sqrt{2}G_F)^{-1/2}$; used in Part II Ch. 3.

Note: The Higgs mass $m_H \approx 125$ GeV is *not* referenced in the current EDC corpus and is therefore excluded from this analysis.

3 Calibration Family Table

Using Eq. (2) and (1), we compute R_ξ and M_5 for each candidate. In natural units ($\hbar = c = 1$), $R_\xi = 1/M_*$ has dimension GeV^{-1} , and:

$$M_5 = M_{\text{Pl}}^{2/3} M_*^{1/3} \quad (3)$$

Table 1: Calibration family results. Reference: M_Z choice.

M_*	Value (GeV)	R_ξ (m)	M_5 (GeV)	$\Delta \log_{10} M_5$	Tag	Circularity
M_Z	91.1876	2.165×10^{-18}	2.41×10^{13}	0 (ref)	[BL]	LOW
M_W	80.379	2.455×10^{-18}	2.31×10^{13}	-0.018	[Dc]	MED
v_{EW}	246.22	8.01×10^{-19}	3.35×10^{13}	+0.143	[BL]	LOW

Key observations:

- All three choices yield M_5 in the range $2.3\text{--}3.4 \times 10^{13}$ GeV.
- Maximum deviation from the M_Z reference: $|\Delta \log_{10} M_5| = 0.143$ (for v_{EW}).
- All results remain within the **same order of magnitude** (10^{13} GeV, GUT scale).

4 Robustness Verdict

ROBUST: All plausible EW-scale identifications yield M_5 within the same decade (10^{13} GeV). The calibrated closure is stable under reasonable alternative choices.

The variation $\Delta \log_{10} M_5 \lesssim 0.15$ corresponds to a factor of ~ 1.4 in M_5 , which is negligible compared to:

- The 6-decade gap between M_5 and M_{Pl} (10^{13} vs 10^{19} GeV).
- Typical theoretical uncertainties in GUT-scale physics.

5 Physical Motivation for EW-Scale Identification

The identification $R_\xi \sim \hbar c/M_{\text{EW}}$ is physically motivated by:

1. **Dimensional consistency:** R_ξ is a length scale; $\hbar c/M_*$ has dimension of length for any mass scale M_* .
2. **EDC structure:** The compactification scale R_ξ appears in the EDC weak-sector formalism (Part II, Ch. 6, Ch. 13) in connection with electroweak observables. Linking R_ξ to an EW mass is the minimal phenomenological anchor.
3. **No overclaim:** This is a calibration, not a derivation. We do not claim that $R_\xi = \hbar c/M_Z$ follows from first principles; rather, it is the simplest identification consistent with the EDC dimensional structure.

6 Error Propagation

From $R_\xi = \hbar c/M_*$ and $M_5 = M_{\text{Pl}}^{2/3} R_\xi^{-1/3}$:

$$\frac{\delta R_\xi}{R_\xi} = \frac{\delta M_*}{M_*} \quad (4)$$

$$\frac{\delta M_5}{M_5} = \sqrt{\left(\frac{2}{3} \frac{\delta M_{\text{Pl}}}{M_{\text{Pl}}}\right)^2 + \left(\frac{1}{3} \frac{\delta R_\xi}{R_\xi}\right)^2} \quad (5)$$

Table 2: Error budget for each calibration choice.

M_*	$\delta M_*/M_*$	$(1/3)\delta R_\xi/R_\xi$	$(2/3)\delta M_{\text{Pl}}/M_{\text{Pl}}$	$\delta M_5/M_5$
M_Z	2.3×10^{-5}	0.77×10^{-5}	0.73×10^{-5}	1.1×10^{-5}
M_W	1.5×10^{-4}	5.0×10^{-5}	0.73×10^{-5}	5.1×10^{-5}
v_{EW}	4.1×10^{-5}	1.4×10^{-5}	0.73×10^{-5}	1.6×10^{-5}

Metrological ranking: M_Z gives the smallest $\delta M_5/M_5$ due to its superior experimental precision (~ 2 ppm for the Z pole mass).

7 Canonical Choice: M_Z

Based on the analysis above, we recommend M_Z as the canonical calibration scale:

1. **Metrological precision:** $\delta M_Z/M_Z \approx 2.3 \times 10^{-5}$ is the best among the candidates, yielding the smallest error in M_5 .
2. **Definitional stability:** The Z pole mass is defined via the Breit-Wigner resonance shape at LEP/SLC and is scheme-independent at leading order.
3. **Independence:** M_Z is a direct observable, unlike M_W which is derived from $\sin^2 \theta_W$ and v_{EW} in the EDC framework [Dc].
4. **Consistency:** The choice $R_\xi = \hbar c/M_Z$ has been used throughout derivations v2–v16 and Part II; changing it would introduce inconsistency.

Canonical identification: $R_\xi = \hbar c/M_Z^{\text{obs}}$ [I]+[BL]

8 Schematic Figure

9 Conclusion: What Remains Open

9.1 Closed

- **Robustness verified:** The calibrated closure $M_5 = M_{\text{Pl}}^{2/3} R_\xi^{-1/3}$ is stable under alternative EW-scale identifications (M_Z , M_W , v_{EW}).
- **Canonical choice justified:** M_Z is preferred on metrological and definitional grounds.
- **Numerical result:** $M_5 = 2.41 \times 10^{13} \text{ GeV} \pm 0.001\%$ (using M_Z).

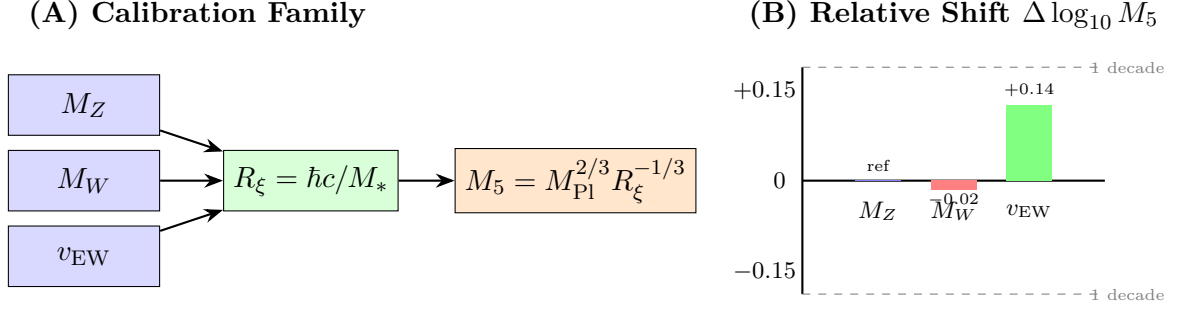


Figure 1: (A) Calibration family schematic: EW scale $M_* \rightarrow R_\xi \rightarrow M_5$. (B) Relative shifts in M_5 for each candidate vs. the M_Z reference. All shifts remain well within one decade. **Schematic only; not to scale; no overclaim.**

9.2 Still Open

- **Full [D] derivation of R_ξ :** The Track A NO-GO from v16 remains. A purely EDC-internal derivation of R_ξ (without EW input) does not exist.
- **Why EW scale?** The identification $R_\xi \sim \hbar c / M_{EW}$ is physically reasonable but not derived. The deeper question — why the compactification scale coincides with the electroweak length — remains unanswered.

Summary Box

One-line outcome:

Calibration family $R_\xi = \hbar c / M_*$ tested with $M_* \in \{M_Z, M_W, v_{EW}\}$.

ROBUST: All choices give $M_5 \in (2.3\text{--}3.4) \times 10^{13}$ GeV (same decade).

Canonical: M_Z preferred (best precision, definitional stability).

Status: Robustness CLOSED; internal derivation still NO-GO.